From Intake to Outcome: Exploring Return-to-Owner Dynamics at HSHV

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Introduction

The rate at which animals brought to the Humane Society of Huron Valley (HSHV) are returned to their owner (RTO) has fallen in recent years, with particularly pronounced declines observed among dogs. This trend raises important questions about potential structural or behavioral changes affecting animal recovery efforts in the region. While some RTO rate fluctuations due to variation in intake volume or seasonal patterns are expected, a sustained decrease suggests possible shifts in community engagement, shelter protocols, or other underlying factors.

Understanding the determinants of RTO rates is critical not only for optimizing shelter operations and animal welfare outcomes, but also for informing public policy and community outreach strategies. Species differences – particularly between dogs and cats – may reflect distinct ownership patterns, microchipping practices, or attitudes toward stray animals. Furthermore, the COVID-19 pandemic may have altered both animal intake patterns and reclaim behaviors, making it necessary to differentiate between temporal effects and broader contextual changes. The role of geography also warrants attention: the location where animals are found could reflect differences in neighborhood infrastructure, socioeconomic status, or proximity to shelter services, all of which may influence the likelihood of an animal being reclaimed.

Using HSHV's database on animal intake and RTO status, STATCOM's analysis aims to identify associations between characteristics of found animals and their eventual RTO outcome. In particular, this report explores the following questions:

1) Are there significant differences in RTO rates between dogs and cats? If so, what factors (e.g., species, location found, or reasons for bringing in animals) are associated with these differences?
2) What are the temporal trends in the RTO rate for dogs and cats over the past 6 years? Are there seasonal variations in the number of animals brought to HSHV and their RTO rates, and do these trends differ pre- and post-pandemic?
3) How do the geographic locations of found animals correlate with specific neighborhoods or shelter proximity? Are there identifiable clusters of high or low RTO rates?
4) Is the location of the finder associated with the likelihood of an animal being reunited with its owner?

I. Are there significant differences in RTO rates between dogs and cats? If so, what factors (e.g., species, location found, or reasons for bringing in animals) are associated with these differences?

Animals in the dataset were categorized into three species groups: dog, cat, and other. Table 1 summarizes the mean time-to-outcome and total counts of animals brought to HSHV, stratified by return-to-owner status, outcome type, and species.

Among animals not reclaimed by their owners, adoption is the most common outcome for both dogs and cats. Notably, mean time-to-outcome in this group is considerably higher than for any RTO pathway, with average durations of 22.14 days for cats (n = 3914) and 22.55 days for dogs (n = 859). These extended timeframes likely reflect the time-intensive nature of the adoption process, which includes medical and behavioral evaluation, placement in foster or adoption-ready status, and the matching of animals with prospective adopters. The disproportionately high number of cats adopted may reflect differences in stray ownership norms or identification practices relative to dogs.

In contrast, animals that were successfully reclaimed exhibit much shorter times-to-outcome. Dogs and cats reclaimed via microchip had mean times of 0.52 and 0.55 days, respectively, highlighting the success and importance of microchipping initiatives as a means of pet reunification. Similarly, dogs reclaimed through owner visit or ID tag were typically returned within 0.92 days (n = 279), demonstrating the value of visible identifiers in the field. However, less direct methods – such as lost reports or social media – appear to be less efficient and less frequently used. Cats and dogs reclaimed via lost reports had longer mean times (1.65 and 0.78 days, respectively), and only 17 dogs were reclaimed through social media (1.29 days). These findings suggest either logistical inefficiencies in matching intake data to owner-submitted lost reports or low public engagement with these mechanisms. Despite their potential, these channels currently appear underutilized relative to their capacity to facilitate timely returns. As shown in Figure 1, the probability of RTO drops substantially beyond 24-48 hours post-intake – highlighting the importance of utilizing various identification and outreach modalities to rapidly reunite owners with their lost pets.

On average, cats are reclaimed less frequently and take longer to be returned than dogs. This species gap is likely influenced by differences in identification and traceability. For dogs, more specific breed labeling may reflect either more detailed intake information or a stronger pre-existing connection to an owner, ultimately leading to more prompt reunification. As seen in Table 2, five most frequently recorded missing dog breeds were Pit Bull Terrier (n = 538), Labrador Retriever (n = 119), Shepherd (n = 88), Hound (n = 66), and Terrier (n = 62). In contrast, over 90% of missing cats were broadly classified as Domestic Shorthair, Longhair, or Mediumhair, with little breed-level distinction. This lack of specificity may reflect a lower prevalence of owner-supplied information or fewer identifying features recorded at intake, reducing the chances of matching an animal to a lost report or microchip registration. The absence of a comparable "named breed" structure in cats may thus contribute to both lower RTO rates and longer time-to-outcome.

These findings support targeted recommendations: HSHV should continue to prioritize microchipping and visible identification, particularly at the point of adoption. Moreover, education efforts might be focused on promoting these identification methods for cats, where reclaim rates and times are lagging. Finally, a closer examination of the lost report process and social media integration may reveal opportunities to improve system responsiveness for animals not immediately identifiable at intake.

Table 1: Mean time-to-outcome and count by outcome type, species, and reclaim status for animals recorded in the HSHV database.

Reclaim Status	Outcome Type	Species	Mean Time-to-Outcome (days)	Count
Not Reunited	Adoption	Cat	22.14	3914
		Dog	22.55	859
		Other	19.89	139
	Transfer Out	Cat	1.89	18
Reunited	Lost Report	Cat	1.65	51
		Dog	0.78	81
	Microchip	Cat	0.55	255
		Dog	0.52	190
	Other Tag	Dog	0.00	17
	Owner Visit/ID	Cat	2.70	130
		Dog	0.92	279
	Social Media	Dog	1.29	17

Table 2: Top missing breeds among dogs and cats brought to HSHV.

Species	Breed	Count
Dog	Pit Bull Terrier	538
	Labrador Retriever	119
	Shepherd	88
	Hound	66
	Terrier	62
Cat	Domestic Shorthair	5,782
	Domestic Longhair	700
	Domestic Mediumhair	342

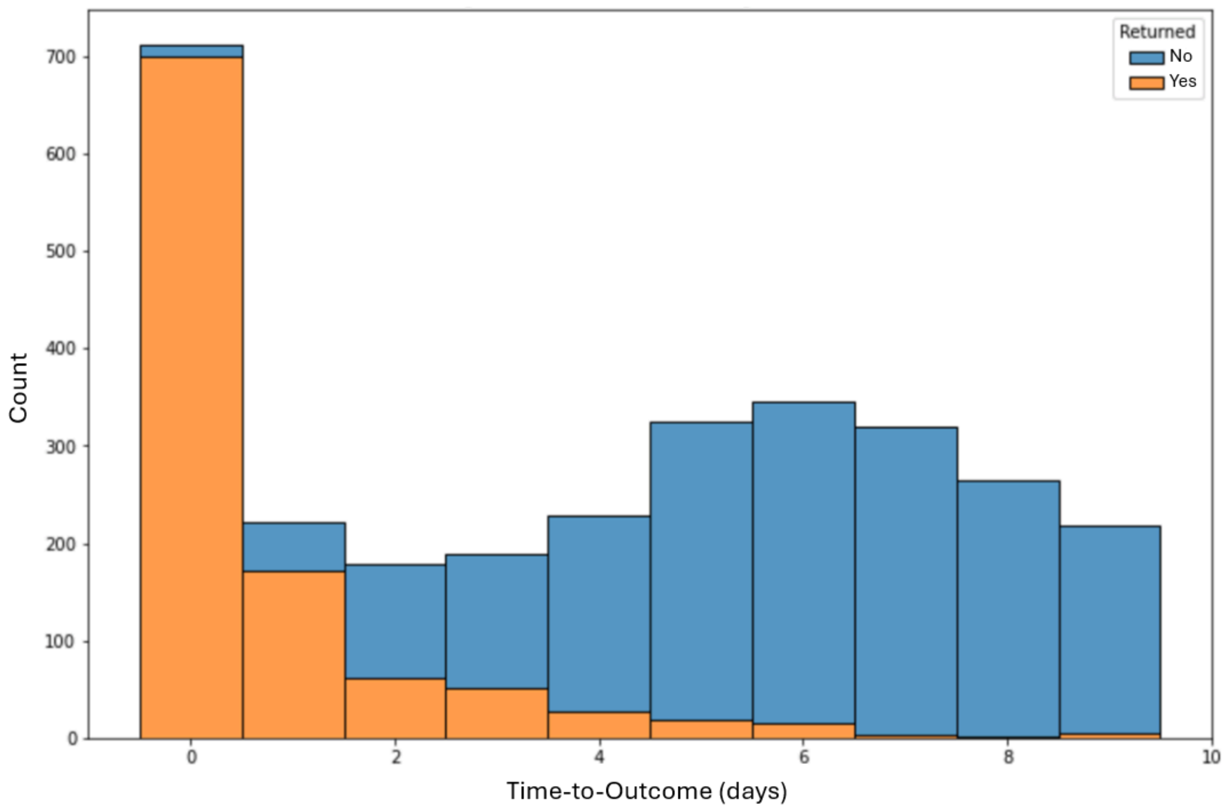


Figure 1: Distribution of time-to-outcome (in days) for animals brought to HSHV, stratified by return-to-owner (RTO) status. Most animals that were reclaimed (orange) were returned on the day of intake or within 48 hours, while non-reclaimed animals (blue) exhibited a wider, left-skewed distribution of outcome times – with most outcomes occurring after several days.

II. What are the temporal trends in the RTO rate for dogs and cats over the past 6 years? Are there seasonal variations in the number of animals brought to HSHV and their RTO rates, and do these trends differ pre- and post-pandemic?

To evaluate changes in reclaim outcomes over time, we examined RTO rate trends by species from 2019 to 2024, with emphasis on seasonal patterns and the impact of the COVID-19 pandemic. Understanding how intake and RTO rates fluctuate across time can inform resource planning, staffing, and targeted interventions.

We first examined RTO rate trends aggregated across all species (Figure 2). This analysis revealed consistent seasonal trends in the number of missing animals brought to HSHV across all six years. In each year, intake volumes rise sharply beginning in spring and peak during the summer months – typically between June and August – before declining again in the fall and winter. This pattern is evident both pre-pandemic (2019) and post-pandemic (2021-2024), suggesting that seasonal variation in animal intake is stable over time and largely unaffected by the pandemic's disruptions.

To investigate whether seasonal variation in missing animal intakes is driven by specific species, a stratified time series analysis was performed. Figure 3 displays monthly intake counts at HSHV from 2019 to 2024 by species. Cats show pronounced seasonal trends, with intake volumes consistently exceeding 100 animals per month during summer, suggesting a strong seasonal effect; dogs exhibit a similar but attenuated pattern. Figure 4 further stratifies these trends by RTO status. Notably, seasonal variability among cats is almost entirely attributable to animals that are not reclaimed; intake of returned cats remains stable year-round. This divergence implies that seasonal surges in missing cats are less likely to result in reunification, highlighting the need for targeted interventions during peak months to improve RTO rates among this group.

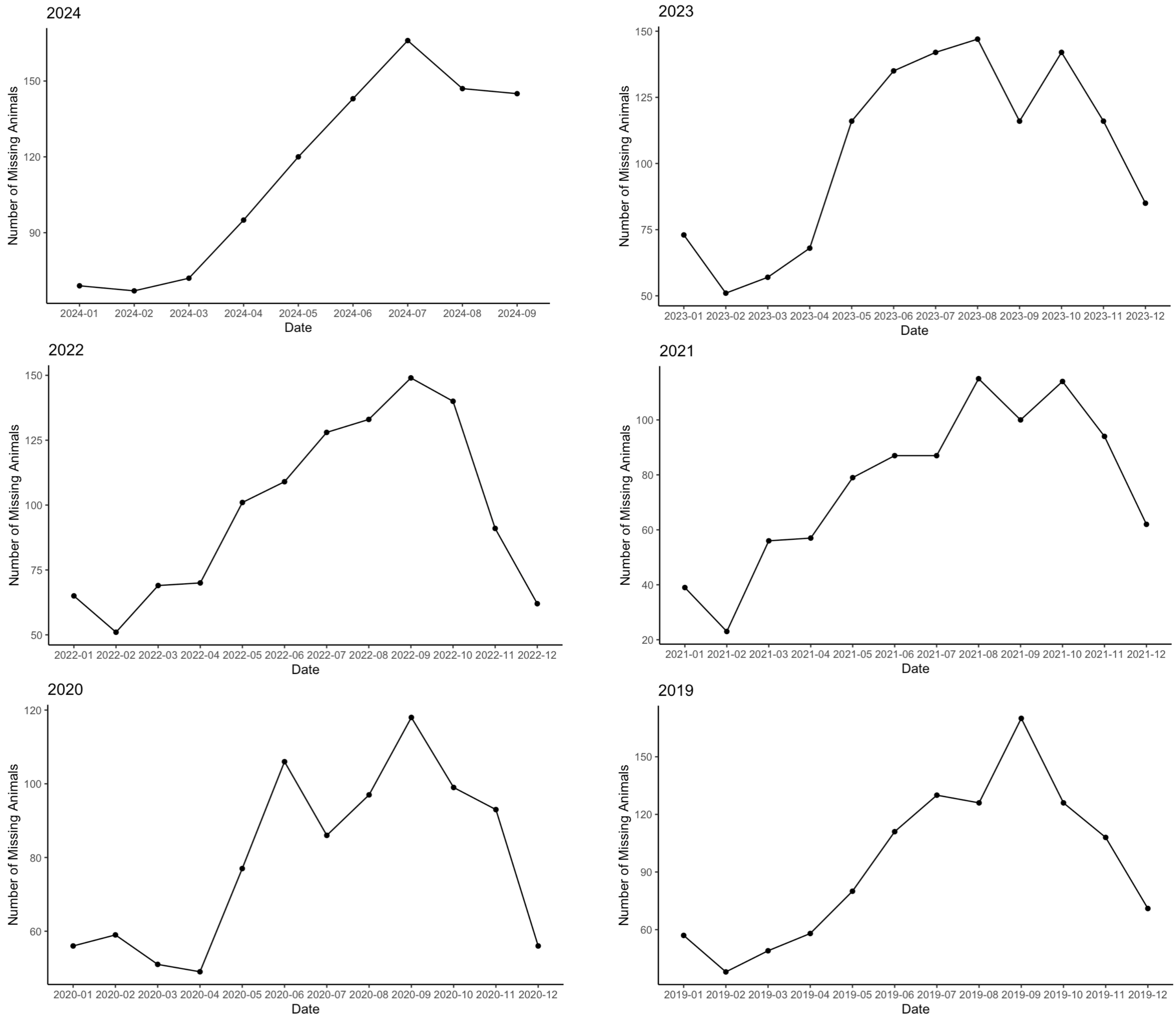


Figure 2: Monthly counts of missing animals brought to HSHV from 2019 to 2024. Each panel represents one calendar year and displays the temporal trend in missing animal intakes, showing consistent seasonal variation with peak activity during summer months and lower volumes in winter.

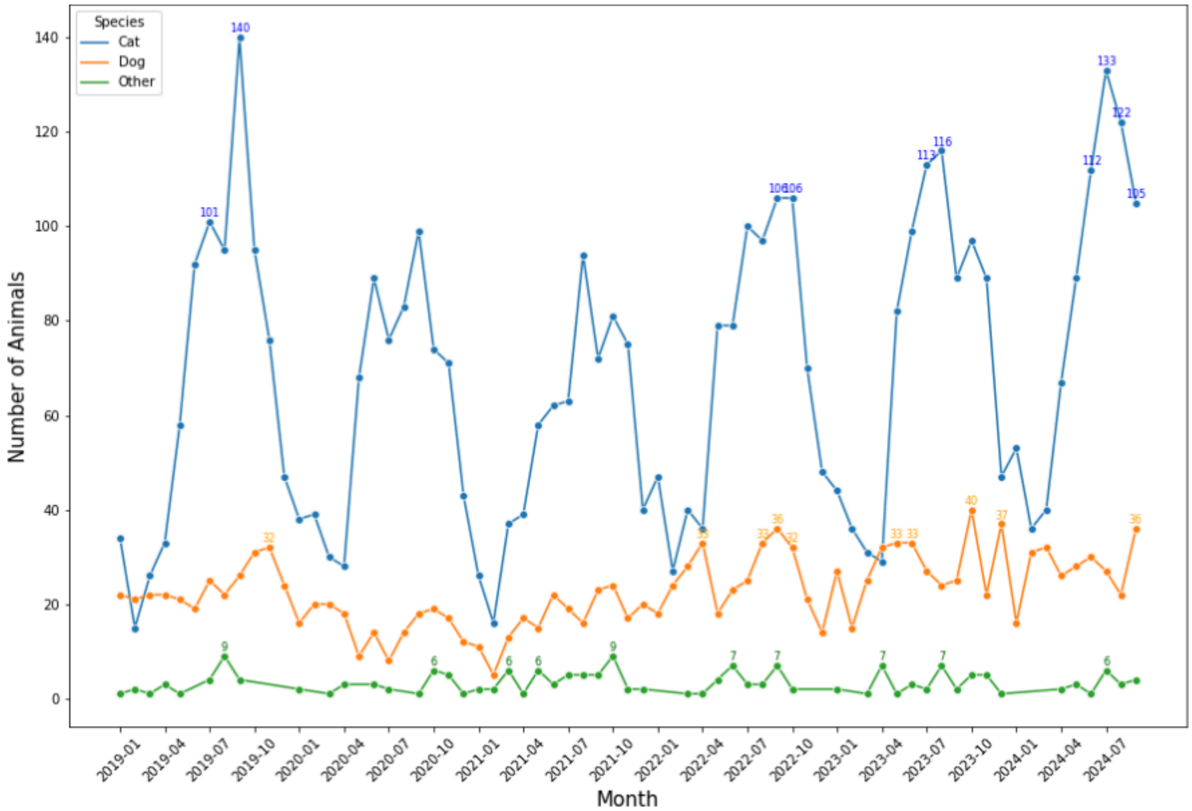


Figure 3: Monthly counts of animal intakes at HSHV from 2019 to 2024, stratified by species. Cats showed strong seasonal fluctuations with consistent peaks in intake during summer months, while dog intakes were more stable over time with modest variability. Intake of other species remained low and relatively constant throughout the period.

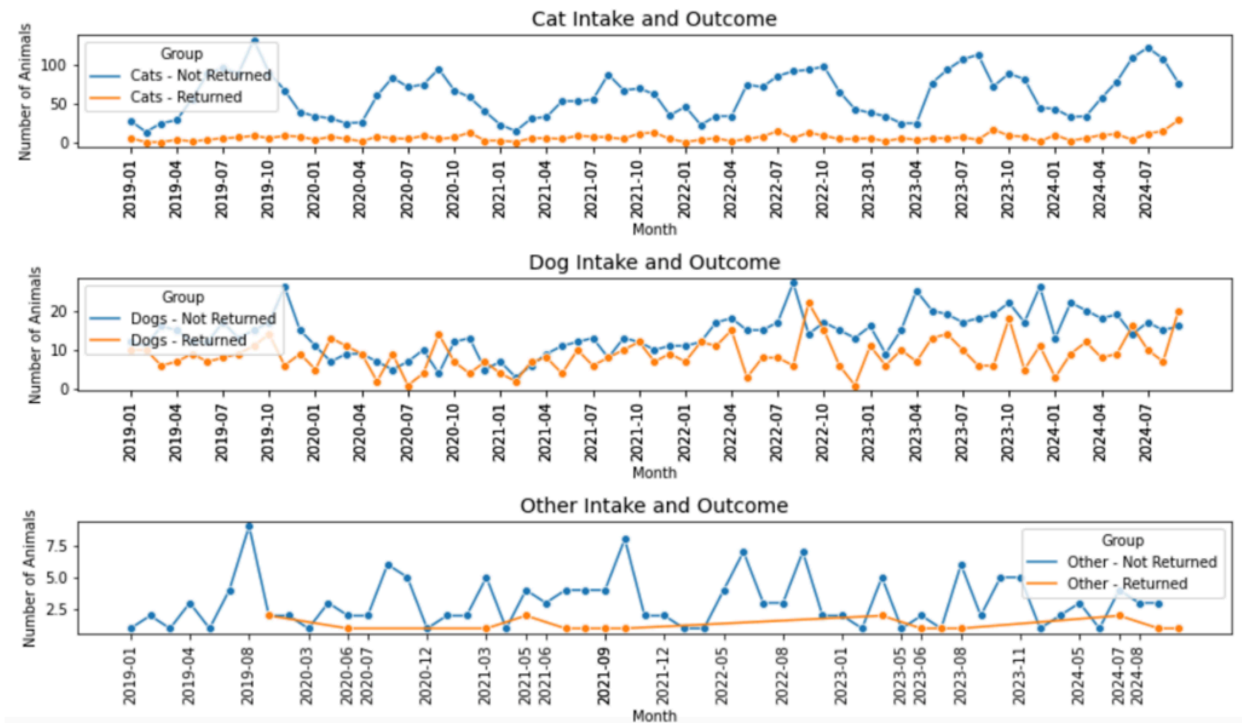


Figure 4: Monthly counts of animal intakes at HSHV from 2019 to 2024, stratified by species and RTO outcome (reclaimed = orange, not reclaimed = blue). The proportion of animals not returned remained high across all species, with cats having pronounced summer peaks in non-returned intakes – suggesting a seasonal intake burden not matched by corresponding RTO rates.

III. How do the geographic locations of found animals correlate with specific neighborhoods or shelter proximity? Are there identifiable clusters of high or low RTO rates?

To examine the spatial distribution of missing animals and its relationship to return-to-owner (RTO) rates, intake locations were geocoded and mapped. Locations of found animals were provided as cross-streets (pairs of street names in reference to the nearest intersection) in the vast majority of cases. Google's geocoding API was used to convert cross-streets into latitude and longitude, which were needed as input for mapping software.

The geocoding API was unable to precisely identify some addresses, resulting in non-specific locations. Among these, 13 addresses were labeled only as "Michigan, USA" and were manually corrected. Additionally, any addresses associated with animals found at HSHV (n = 94) or outside the state (n = 621) were excluded from the dataset. To ensure granularity of the addresses for analysis, rows lacking zip codes were filtered further. Specifically, addresses without detailed information – characterized by the absence of keywords such as "Ave," "Rd," "Dr," "Trail," "St," "Pkwy," "&," "Park," or "Lake" – were removed, excluding a total of 1,663 observations. Duplicates were then identified and removed based on key fields: 'Intake Date,' 'Species,' 'Primary Breed,' 'Location Found,' and 'Returned to Address.' A total of 3,674 duplicates were identified, and only the first instance of each duplicate was retained. This process reduced the dataset to 6,424 unique observations.

To translate coordinate-level information into a more interpretable visualization, we utilized Streamlit – which is commonly used to develop interactive data apps – in order to create heat maps and hex plots displaying the spatial distribution of found animals overlaid on Michigan. Figure 5 presents Streamlit-generated spatial heatmaps of abandoned dog locations from the

HSHV database. The left panel (a) highlights spatial clustering, while the right panel (b) overlays individual outcomes – specifically whether a dog was returned (red) or not (blue). For visual clarity, groups of animals near each other have been combined into clusters, each labeled with the number of animals being represented. The color of the clusters represents frequencies, ranging from green (low) to orange (high). By clicking the layer control in the top right corner, users can view data from different years. These visualizations suggest geographically heterogeneous RTO rates, with some areas potentially showing disproportionately low reunification despite high intake volume. However, the distribution of abandoned dogs appears to line up with overall population density within and around Ann Arbor and Ypsilanti.

To further quantify these patterns, Figure 6 displays a hexbin plot, where hexagon height denotes the number of dogs found in a given area and color intensity reflects the proportion not returned. Darker red areas with tall bars represent high-volume, low-RTO zones – indicating neighborhoods with both elevated intake frequency and poor reunification outcomes. In contrast, lighter or shorter bars denote either fewer dogs found or higher rates of successful return. These spatial trends suggest opportunities for geographically targeted interventions such as microchipping campaigns, public awareness efforts, or improved access to reporting infrastructure in low-RTO zones.

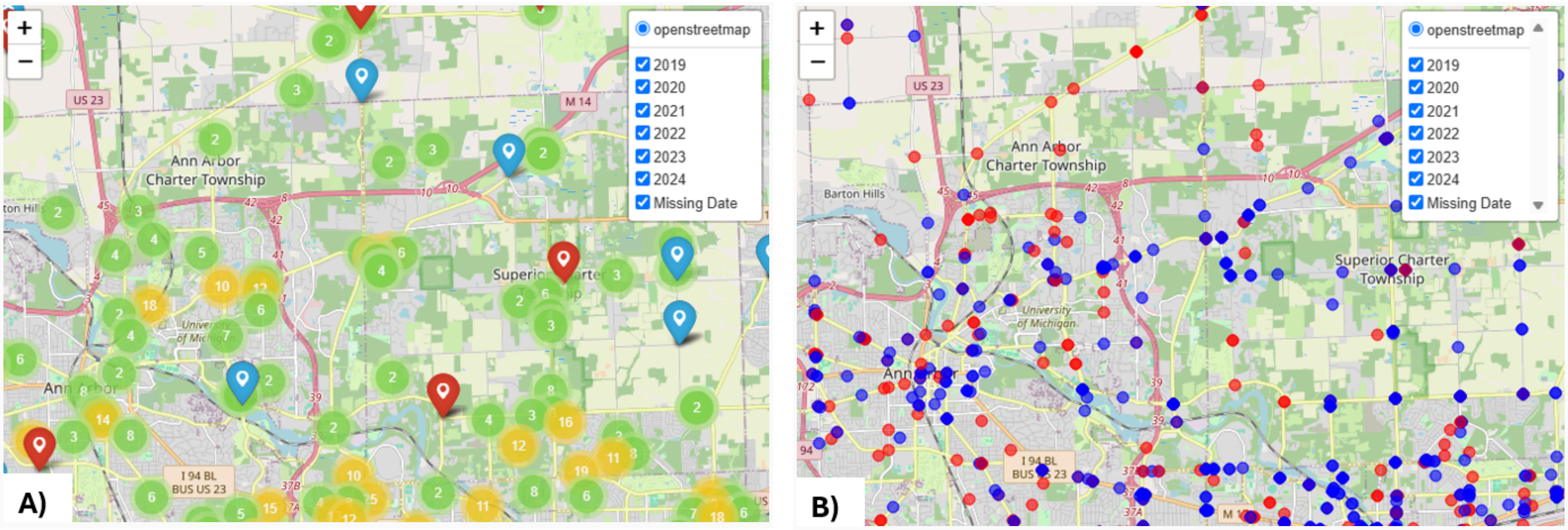


Figure 5: Streamlit implementations of spatial heatmap showcasing dogs from HSHV database a) with clustering and b) without clustering. Blue points represent dogs not reunited with their owner, while red points represent those which were reunited.

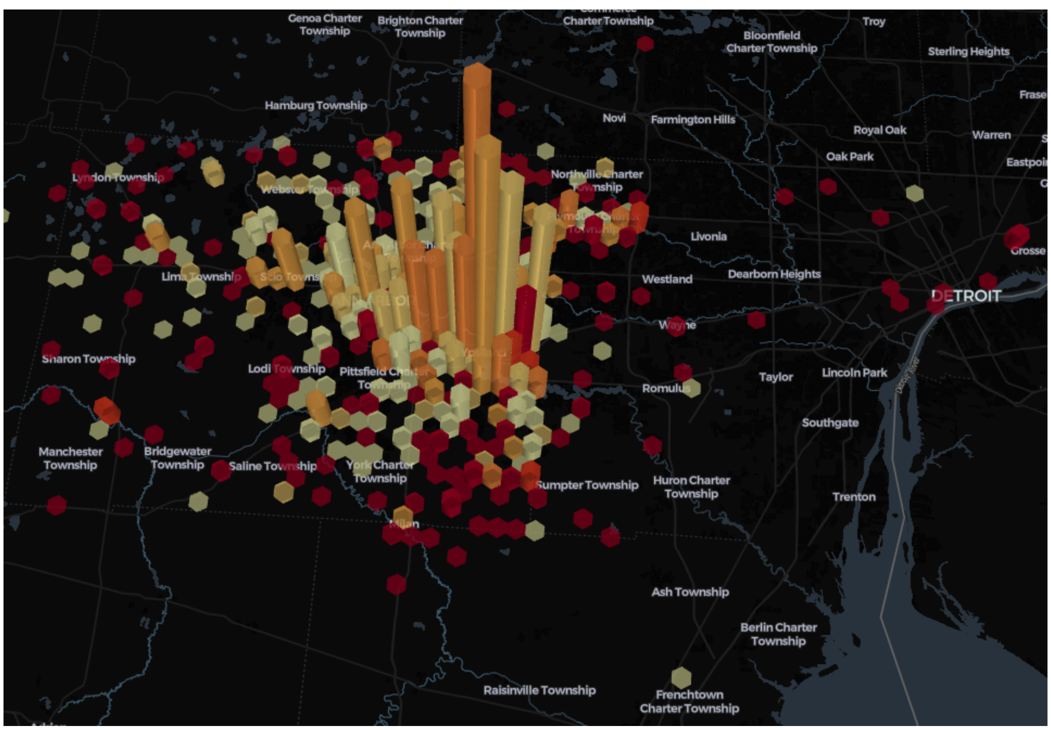
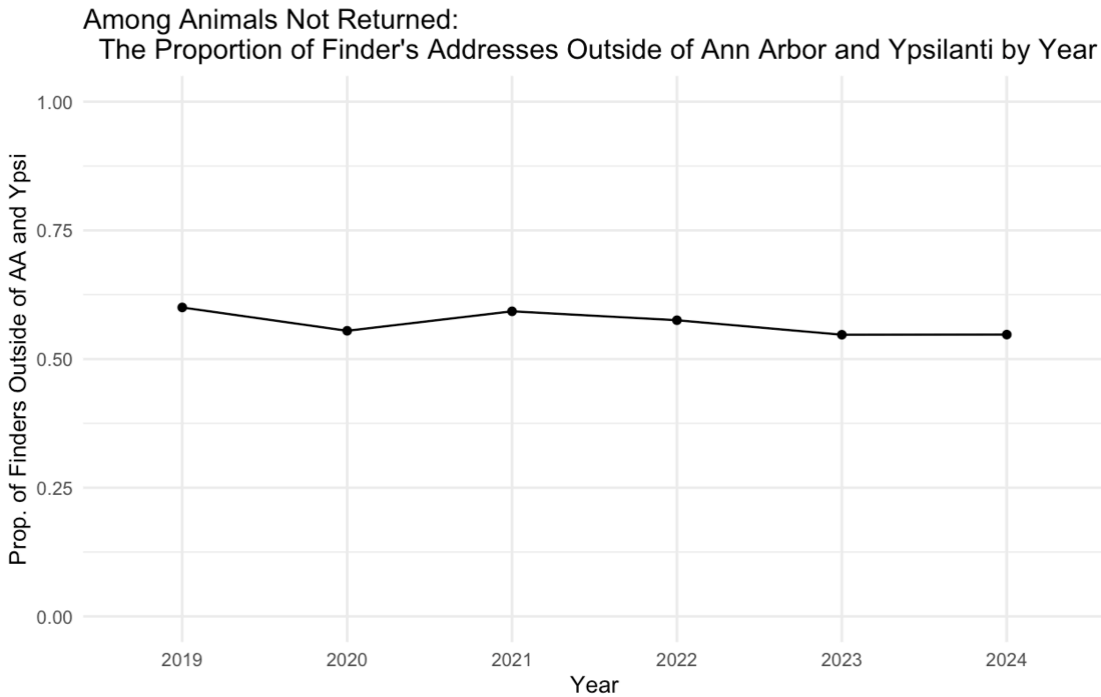


Figure 6: Hexagon plot depicting counts and proportions of dogs not returned in a given area. Taller bars correspond to a higher number of dogs found in the area, with a darker red color indicating fewer returns to owners of said dogs. Areas of highest interest are indicated by areas with both tall and red bars.

IV. Is the location of the finder associated with the likelihood of an animal being reunited with its owner?

Owner reunification may be influenced not only by animal-level factors but also by geographic or contextual variables, including where and by whom the animal was found. To explore whether finder location correlates with return-to-owner (RTO) outcomes, we stratified animals by RTO status and examined whether the finder's address fell outside of Ann Arbor and Ypsilanti – the two municipalities closest to HSHV and the shelter's primary catchment area.

Figure 7 presents the annual proportion of finders residing outside Ann Arbor and Ypsilanti from 2019 to 2024, stratified by RTO status. Among non-reclaimed animals (top panel), roughly 55-60% were found by individuals living outside these cities, with this proportion remaining relatively stable across years. In contrast, among successfully reunited animals (bottom panel), approximately 35-40% were found by individuals from outside the core cities. This spatial disparity suggests that animals found by individuals residing farther from the shelter's geographic center may face barriers to reunification, possibly due to reduced shelter proximity, limited knowledge of HSHV services, or lower likelihood of microchipping/reporting in surrounding regions.



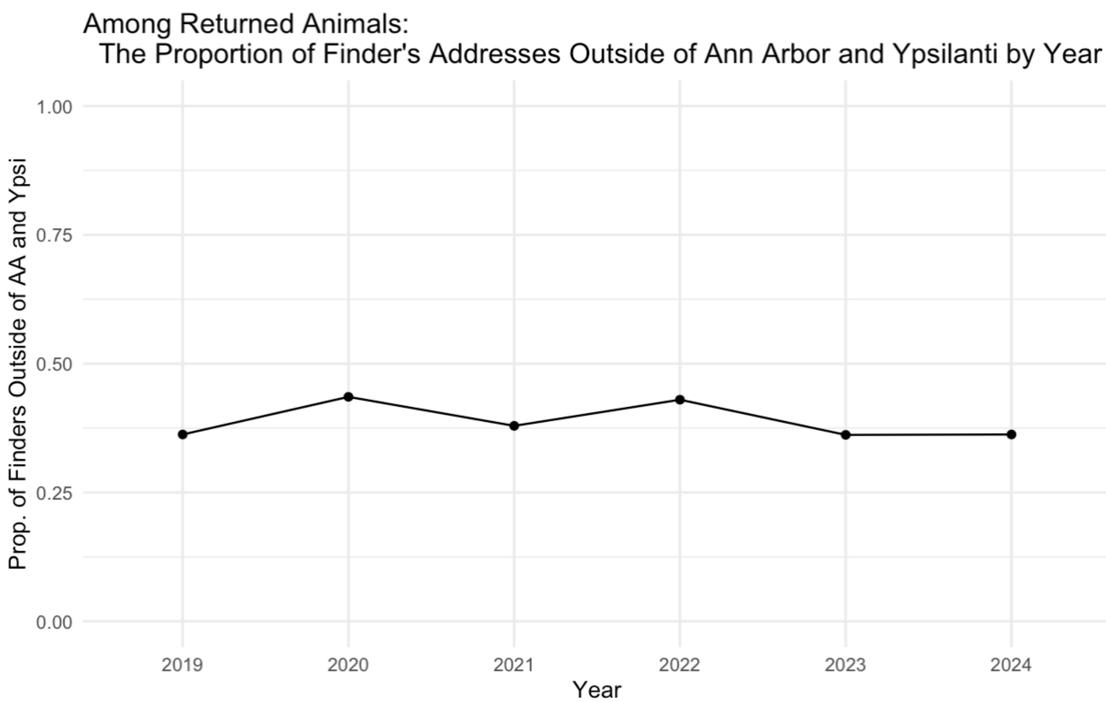


Figure 7: Annual proportion of finders residing outside Ann Arbor and Ypsilanti, stratified by return-to-owner status (2019–2024). Non-reclaimed animals were more often found by individuals outside the core shelter cities, with proportions remaining stable over time.

Credits

Anran: Creation of Folium heatmaps and Streamlit app architecture
Helen: Exploratory data analysis, investigation of relationship between far-away finder addresses and RTO rate among dogs
Isha: Creation of geocoding script, data deduplication/quality control filtering, and finalization of dataset for subsequent analyses
Kevin: Hexagon map plot coding
Zach: Exploratory data analysis, dog RTO analysis and calculation feature of Streamlit app
Kirill: Application of geocoding script to HSHV database, team organization and report writing/formatting
Sophia: Statistical analysis planning, Streamlit app architecture, and heatmap coding